



Brief Communication

Azithromycin resistance and its molecular characteristics in *Neisseria gonorrhoeae* isolates from a tertiary care centre in North IndiaApoorva Challa^a, Neeraj Mahajan^b, Seema Sood^{b,*}, Arti Kapil^b, Bimal Kumar Das^b, Vishnubhatla Sreenivas^c, Somesh Gupta^a^a Department of Dermatology and Venereology, All India Institute of Medical Sciences, New Delhi 110029, India^b Department of Microbiology, All India Institute of Medical Sciences, New Delhi 110029, India^c Department of Biostatistics, All India Institute of Medical Sciences, New Delhi 110029, India

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ABSTRACT

Treatment guidelines for management of uncomplicated gonorrhoeae have been recently modified owing to alarming upsurge in azithromycin resistance. This study investigated the prevalence and genetic determinants of gonococcal azithromycin resistance in India. Four (5.7%) of 70 gonococcal isolates were resistant to azithromycin. Of 16 isolates investigated for molecular mechanisms of resistance, 13 (81.3%) and 6 (37.5%) isolates exhibited mutations in coding and promoter regions of *mtrR* gene, respectively. However, *ermA*, *ermB* and *ermC* genes or mutations in *rrl* gene were absent in all isolates. Azithromycin resistance is low in India posing no immediate threat to use of dual-therapy for syndromic management.

1. Introduction

Gonorrhoeae continues to be a significant public health concern due to increasing global burden and emerging resistance to antimicrobials. In response to an alarming upsurge in azithromycin resistance, CDC now recommends the use of a single intramuscular dose of ceftriaxone (500 mg) for treatment of uncomplicated gonorrhoeae and empirical co-treatment with oral doxycycline for *Chlamydia trachomatis* [1]. The latest World Health Organisation (WHO) global surveillance of gonococcal AMR has identified azithromycin resistant isolates in 51 (84%) of 61 reporting countries, wherein, the WHO- South East Asian Region (SEAR) reported resistance in 2 (50%) countries [2]. In India, gonococcal azithromycin resistance has shown an upward trend, although, the number of isolates being investigated has drastically reduced in the last decade, underscoring the need for continued nation-wide surveillance. A non-significant ($p > 0.05$) increase in azithromycin resistance was noted between 2002–2006 (0.8% of 133 isolates) and 2007–2012 (1.5% of 261 isolates) at the Apex STD Centre in India [3]. Subsequent report by Kulkarni et al. in 2016, identified resistance in 6 (4.8%) of the 124 isolates included from various centres across the country [4]. More recently, 3 (9.4%) of the 32 isolates were found to be azithromycin resistant [2]. The decrease in azithromycin susceptibility has been attributed to the multiple molecular mechanisms of resistance identified in *Neisseria*

gonorrhoeae including alteration of ribosomal target by either mutation in peptidyltransferase loop of 23S rRNA (*rrl*) gene or methylation of rRNA methylase (*erm*) genes and over expression of efflux pump due to mutations in promoter and coding regions of *mtrR* gene [5]. Although mutations of the *rrl* and *mtrR* genes are most commonly reported mechanisms, trends in antimicrobial susceptibility patterns have shown geographical variations. The present study was undertaken to investigate the prevalence and genetic determinants of azithromycin resistance in gonorrhoeae in a tertiary care centre in North India.

2. Materials and methods

Seventy gonococcal isolates obtained between 2013 and 2017 were revived and minimum inhibitory concentration (MICs) for azithromycin and ceftriaxone were determined using E-test (Biomerieux, France) and interpreted by calibrated dichotomous sensitivity (CDS) technique [6]. Ethical approval was obtained from the Institute Ethics Committee, All India Institute of Medical Sciences, New Delhi and informed consent was obtained from all participants. The breakpoints for susceptibility and resistance to azithromycin were $<1.0 \mu\text{g/ml}$ and $\geq 1.0 \mu\text{g/ml}$, respectively and susceptibility and decreased susceptibility to cephalosporin were $<0.06 \mu\text{g/ml}$ and $0.06\text{--}0.25 \mu\text{g/ml}$, respectively. Additionally, all isolates were tested for β -lactamase production by the chromogenic cephalosporin

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Table 1β-lactamase production, minimum inhibitory concentrations (MICs) of ceftriaxone and azithromycin and *mtrR* gene mutations in 16 gonococcal isolates.

Strain No	β-lactamase	Ceftriaxone MIC (μg/ml)	Azithromycin MIC (μg/ml)	Sequencing of <i>mtrR</i>							
				<i>mtrR</i> promoter	Mutations in coding region						
					L33V	A39T	L47P	G45D	C66Y	H105Y	D171A
B15	Negative	0.012 (S)	0.50 (S)	No mutation	X	X	X	✓	X	X	X
L09	Positive	<0.002 (S)	0.38 (S)	No mutation	X	X	X	X	X	X	X
L22	Negative	0.003 (S)	2.0 (R)	No mutation	X	X	✓	✓	X	X	X
L29	Positive	0.002 (S)	0.25 (S)	No mutation	X	X	X	X	X	X	X
R5	Negative	0.023 (S)	2.0(R)	No mutation	X	✓	✓	X	X	X	X
R7	Negative	0.064 (DS)	0.75 (S)	A deletion	X	X	X	X	X	✓	X
R10	Negative	0.064 (DS)	0.125 (S)	A deletion	X	X	X	X	X	✓	X
R33	Positive	0.064 (DS)	0.125 (S)	A deletion	X	X	X	X	X	✓	X
R167	Negative	0.002 (S)	0.094 (S)	No mutation	X	✓	X	X	✓	X	X
SP146	Negative	0.125 (DS)	0.25 (S)	A deletion	X	X	X	X	X	✓	X
SP243	Negative	0.016 (S)	1.0 (R)	A deletion	X	✓	X	X	X	X	X
SP273	Positive	0.125 (DS)	1.0 (R)	A deletion	X	X	X	X	X	✓	X
SP295	Negative	<0.002 (S)	0.38 (S)	No mutation	X	X	X	✓	X	X	✓
SP596	Positive	<0.002 (S)	<0.016 (S)	No mutation	X	✓	X	X	X	X	X
SP727	Negative	0.032 (S)	0.016 (S)	No mutation	✓	X	X	✓	X	X	X
SP743	Positive	0.003 (S)	0.094 (S)	No mutation	X	✓	X	X	X	X	X

S- Sensitive, DS- Decreased susceptibility, R- Resistant, ✓- mutation present, X- mutation absent.

method using nitrocefin discs (BD, USA) [7]. Sixteen isolates (azithromycin resistant = 4, azithromycin sensitive = 12) were selected to investigate mechanisms of resistance by identification of mutations in 23S rRNA (*rnl*) and *mtrR* genes by DNA sequencing and detection of *ermA*, *ermB* and *ermC* genes by polymerase chain reaction (PCR) based amplification, using primers and methods described previously [8–10]. The following reference strains were used as positive controls: *N. gonorrhoeae* WHO P strain (for *rnl* and *mtrR*) *Staphylococcus aureus* strain ATCC 43300 (for *ermA*), *Escherichia coli* clinical isolate (for *ermB*) & *Staphylococcus aureus* clinical isolate (for *ermC*). DNA was extracted by boiling method [11]. Briefly, colony suspension of *N. gonorrhoeae* isolates were prepared in 100 μl sterile MilliQ, vortexed, boiled for 10 min and subsequently centrifuged at 8000 rpm for 5 min. The supernatant containing the DNA was then separated and stored at –20 °C.

3. Results

Of the 70 gonococcal isolates, 4 (5.7%) isolates were resistant to azithromycin, 13 (20.0%) isolates demonstrated decreased susceptibility to ceftriaxone and 21 (30%) isolates were positive for β-lactamase production. All four azithromycin resistant isolates exhibited low level resistance (MIC = 1–2 μg/ml), 3 were susceptible to ceftriaxone and 1 isolate was also positive for β-lactamase production. Analysis of 23S rRNA gene alleles 1–4 by comparison of sequenced products with *Neisseria gonorrhoeae* NCCP11945 (GenBank accession no: CP001050) revealed that all isolates were wild type. (S1) Also, none of the 16 isolates were positive for the presence of *ermA*, *ermB* or *ermC* genes. On sequencing of the promoter and coding region of *mtrR* and comparison with wild type strain (GenBank accession no: Z25796.1), deletion of adenine (A) in the 13 base-pair inverted repeats of the promoter region was observed in 6 isolates of which 2 were resistant (MIC = 1 μg/ml) to azithromycin (S2 and Table 1). In the coding region of the *mtrR* gene, G45D was seen alone in 1 isolate and in 3 other isolates with additional mutations of L33V, L47P and D171A, respectively. H105Y mutation was seen in 5 isolates, all of which had the deletion in the promoter region of the *mtrR* gene. Similarly, A39T was also noted in 5 isolates; however, 4 of these isolates were wild type (Table 1).

4. Discussion

Mutations in the *mtrR* genes encoding the efflux pump were the only mechanism of azithromycin resistance identified among our study isolates, however, no statistical significance could be established due to low frequency (5.7%) of resistant strains. Gonococcal *mtrR* gene mutations

have been documented in decreased susceptibility to several antibiotics including penicillins, cephalosporins and macrolides [12–14]. Even though global rise in gonococcal azithromycin resistance has prompted revision of treatment guidelines, the level of resistance continues to be low in India posing no immediate threat to previously recommended dual therapy in treating uncomplicated gonorrhoeae [15]. However, emergence of decreased susceptibility to ceftriaxone magnified by injudicious use of azithromycin in treating other infections may imperil the ongoing selective pressure for gonococcal antimicrobial resistance (AMR) [16]. Therefore, continued examination of the evolving epidemiology and molecular mechanisms of drug resistance are warranted. The knowledge of genetic determinations of resistance will promote generation of regional data and aid development of rapid molecular tests for AMR prediction thereby strengthening AMR surveillance. Rapid NAAT based point-of-care tests for *N. gonorrhoeae* and its resistance determinants will favour informed precision medicine strategies thereby preventing onward transmission of gonococcal AMR.

Ethics approval

The study was approved by the Institute Ethics Committee; All India Institute of Medical Sciences and informed consent was obtained from all participants. [IEC/NP-432/2013].

CRedit author statement

Apoorva Challa: Investigation, Data Curation, Writing- Original Draft. **Neeraj Mahajan:** Investigation, Data Curation. **Seema Sood:** Conceptualization, Methodology, Validation, Data Curation, Writing- Review & Editing, Supervision, Project Administration, Fund acquisition. **Arti Kapil:** Conceptualization, Supervision. **Bimal Kumar Das:** Conceptualization, Supervision. **Sreenivas Vishnubhatla:** Formal Analysis, Supervision. **Somesh Gupta:** Methodology, Resources, Supervision.

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Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijmmb.2022.06.002>.

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